LAB 11

Example # 1

intial state → symbol Table is empty

Step1:

sym.insert ('x', INT);

This line says that insert 'x' with type 'INT' in the table as there is no 'x' present in the table.

Step2:

sym.insert ('y', FLOAT);

'y' not in the table, insert 'y' with the Type ~~FLOAT~~ 'FLOAT'

Step3:

sym.insert ('x', INT);

Duplicat detection! As 'x' already exists in the table.

Step4: sym.print();

This print the values present in the symbol table. Output: x : int

y : float

Step 6:

sym.lookup('z');

↳ 'z' not found

It will give an error 'z' not declared

Step5: Type t = sym. lookup ('x');

'x' found and returns int.

Step 7: Final Output

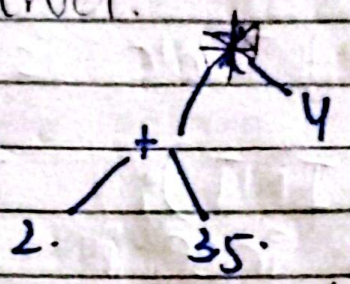
Error: variable 'x' alread declared

x: int y: float

Error: z not declared.

Example 2: Expression (2+3.5)*4

Tree Struct:



Step 1: Check Expr (mul)
operator (*) call left and Right Subtree.

Step2. Check Expr (plus)
plus → left subtre.
node operator + will call its left and right children.

Step3: Check Expr (leaf 2)
no operator, typ INT
return INT.

Step 4: Check Expr (leaf 3.5)
type Float, return → float

Step 5: Evaluate 2+3.5
Float + INT = Float
return float

Step6 check Expr (leaf 4) leaf Node
type INT return int.

Step7: Evaluat (2+3.5)*4.
Float * anytype = Float
return float.

Final:
Expressin type: Float.

Example 3

Type Definition

typedef int Age; → alias of int
typedef int weight; → alias of int

structures

struct Point { int x, int y };
struct Coordinates { int x, y };

The struct names are different, but same structures.

step 1: (Variable Declaration)

Age a = 25;
weight w = 70;

step 2: Print

Output age = 25, weight = 70;

step 3: Structure Intialize

Point P = { 1, 2 }

step 4: Structure Check

auto = (sizeof(poin) == size of (coordi)
Point (two integers) → 8 bytes ∵ 1 int = 4 byte
coordinat (" integer) → 8 bytes, same struct (1)

Step 5: Result.

Point Coordinat have same size (1

Final Message.

C++ uses name equivalen, so they are incompatible

Example 4:

in symbol table empty
(intial state)

Step 1 tc.declare ("x", INT);
Inser "x" withy type INT into symbol table

Step 2: tc.declare ("y", Float);
Now Insert 'y'

Step 3: tc.declar ("name", STRING)
insert "name" into symbol table

Now Assigning

Step 4: tc.checkAssignment ("x", "5", INT);
type = INT
type Match
OUTPUT: ok: x=5

Step 5: tc.check Assignmen ('y', "Float", '3.15');
type Match
output : Ok: y=3.14.

Step 6: tc.check assignment ("x" "name", String)
thistyp ~~this~~ int , this type strin
not Match
output : Typ mismatch cannot assign string to int.

Step 7: tc.checkassignment ("z" "10", INT)
z not declared
type → unknown.
Output: z not declared
Type: Mismatch cannot assign into to unknown